

SI 602: Math for Applied Data Science

Time and Location: Mondays and Wednesdays 9:00 - 10:30am, 1240 Leinweber

Instructor: Alex McLeod (mcleodal@umich.edu)

- **Office Hours:** Wednesday afternoons sometime?

GSIs: Sanzeed Anwar (sanzeed@umich.edu), David Gamba (gamba@umich.edu)

- **Sanzeed's Office Hours:**
- **David's Office Hours:**
- **Monthly Tutorial:**

Course Description:

This course will cover mathematical concepts in linear algebra, probability, statistics, and optimization that are a useful foundation for data science. Students will become comfortable with basic math concepts and how they can be used to solve problems. A few important advanced topics and Python functions will also be covered.

There are five units in the course: Math Basics, Linear Algebra, Probability, Statistics, and Optimization:

- Math Basics covers foundational concepts that will be useful in the other units.
- Linear Algebra covers concepts related to vectors (one column of data) and matrices (many columns of data) that are useful for data science techniques such as Eigendecompositions and OLS Regression.
- Probability covers concepts that allow you to make statements about the chances that a pattern in the data is simply due to randomness (rather than a real relationship in the world) and about the likelihood that competing models gave rise to the data.
- Statistics covers concepts directly related to data science techniques such as the Central Limit Theorem, Hypothesis Testing, OLS Regression, and Logistic Regression.
- Optimization is about finding the maximums and minimums of functions. If you can attach a number to how well the parameters of the model match the data (i.e. if you have an objective function), then you can use optimization techniques to find the optimal parameters.

Recommended Textbooks

There is no required textbook for this course, but “***Essential Math for Data Science***” by Nield covers many of the same topics at a similar level of difficulty. If you feel like you could benefit from another perspective on a topic, I highly recommend you read about it in this book. If you make an O'Reilly account online with your umich email address, then you can access it for free.

There are four other good textbooks that you can use to go deeper into certain topics.

1. *"Linear Algebra and its Applications"* by Lay, McDonald, and Lay
2. *"A First Course in Probability"* by Ross
3. *"Introduction to Econometrics"* by Stock and Watson
4. "Mathematics for Machine Learning" by Deisenroth, Faisal and Ong

Learning Objectives:

1. Math Basics

1. Ability to evaluate expressions that involve set operations.
2. Ability to evaluate simple summations and limits.
3. Ability to evaluate expressions that involve exponents and logarithms.
4. Ability to apply the three main tricks that involve logarithms.
5. Ability to apply the formula for the log approximation to percentage change.
6. Understand the information that a derivative or integral captures.
7. Ability to evaluate simple derivatives or integrals.

2. Linear Algebra

1. Ability to apply basic vector and matrix operations.
2. Familiarity with the connection between the inverse of a function and the inverse of a matrix.
3. Understand that a matrix-vector product results in a linear combination of the column vectors of the matrix.
4. Understand that a matrix-matrix product is a series of matrix-vector products.
5. Ability to solve a system of linear equations or identify that it has zero or infinite solutions.

6. Understand the significance of a matrix having a determinant equal to zero.
7. Understand the relationship between a matrix and its eigenvalue-eigenvector pairs.
8. Understand the formula for an eigendecomposition.
9. Awareness of adjacency matrices and the meaning of one raised to a certain power.

3. Probability

1. Ability to apply the multiplication principle.
2. Ability to apply the formulas for permutations and combinations.
3. Ability to compute the expectation and variance of simple random variables.
4. Ability to apply certain tricks that involve expectation and variance.
5. Ability to answer probability questions that involve the binomial distribution.
6. Ability to represent simple conditional probabilities using Venn diagrams.
7. If provided with a table of joint probabilities, ability to calculate conditional probabilities.
8. Understand the condition for two events (or random variables) to be independent.
9. Ability to answer simple Bayes' rule questions.
10. Ability to answer Bayes' rule questions that involve a series of signals.
11. Understand how Monte Carlo simulations rely on the Law of Large Numbers.
12. Ability to set up Monte Carlo simulations of medium difficulty.

4. Statistics

1. Understand the statement of the CLT.
2. Understand how a hypothesis test relies on the CLT.

3. Be able to do a step-by-step hypothesis test about a population mean.
4. Understand the information conveyed in a confidence interval.
5. Be able to construct a confidence interval.
6. Understand the optimization problem that an OLS or logistic regression solves.
7. Understand how to interpret the results of an OLS or logistic regression.

5. Optimization

1. Understand the definition of a local maximum and minimum and a global maximum and minimum. understand when a global maximum or minimum may not exist.
2. Ability to use the sufficient conditions (i.e. the first and second derivatives) to solve for the local minimums of a simple univariate function.
3. Ability to use the sufficient conditions (i.e. the gradient and the Hessian) to solve for the local minimums of a very simple multivariate function.
4. Understand why gradient descent works.
5. Ability to implement a simple gradient descent algorithm.
6. Awareness of stochastic gradient descent and momentum.
7. Awareness of the main drawbacks of gradient descent.
8. Ability to calculate the likelihood of a parameterization.
9. Awareness of the procedure to find the parameters with the maximum likelihood.
10. Awareness of maximum a posteriori.

Course Requirements:

- There will be a quiz every Wednesday except the first one.
 - That's 13 quizzes.
 - Only your top 10 quizzes will count.
- There will be 7 homeworks.
 - They will be posted on and submitted on Canvas.

- You will write out the answers on paper and upload a pdf.
- They will be returned about one week after the due date.
- The answers will be posted soon after they are returned.
- There will be an in-class midterm. You can use a basic calculator.
- The final exam will be scheduled for exam week. You can use a basic calculator.

Grades

Quizzes: 5%; all quizzes contribute equally to the overall grade.

Homeworks: 45%; all homeworks contribute equally to the overall grade.

Midterm: 20%

Final Exam: 30%

Number Grade to Letter Grade Conversion

97+	93-97	90-93	87-90	83-87	80-83	77-80	73-77	70-73	67-70	63-67	60-63	60-
A+	A	A-	B+	B	B-	C+	C	C-	D+	D	D-	E

Tentative Schedule of Topics Covered

Week 1: Jan 7	Course Overview, Class Activity, Pre-Test
Week 2: Jan 12, 14	Sets (MB), Summations (MB), Limits (MB), Vectors (LA)
Week 3: Jan 21	Exponents and Logs (MB), Span and Basis (LA)
Week 4: Jan 26, 28	Euler's Number (MB), Lin. Indep. (LA), Matrices (LA), Functions (MB)
Week 5: Feb 2, 4	Matrix Mult. (LA), Derivatives (MB), Linear Maps (LA)
Week 6: Feb 9, 11	Integrals (MB), Combinatorics (P)

Week 7: Feb 16, 18	R.V.s (P), Eigenstuff (LA)
Week 8: Feb 23, 25	Expectation (P), Eigendecompositions (LA)
Week 9: Mar 9, 11	Variance (P), The Binomial Dist. and the Normal Dist. (P)
Week 10: Mar 16, 18	SVD (LA), CLT (S), Midterm Exam
Week 11: Mar 23, 25	Hypo Testing (S), Conditional Prob. (P)
Week 12: Mar 30, Apr 1	Independence (P), OLS 1 (S), Bayes' Rule (P)
Week 13: Apr 6, 8	MC Simulations (P), OLS 2 (S)
Week 14: Apr 13, 15	Univariate Functions (O), Logistic Regression (S)
Week 15: Apr 20	Multivariate Functions (O), Gradient Descent (O)
Apr 27, 10am - 1pm	Final Exam

Late Policy

For homeworks submitted between 0-24h after the deadline, a 5 percentage point late penalty will be applied to the grade. For homeworks submitted between 24-48h after the deadline, a 10 percentage point late penalty will be applied to the grade. For homeworks submitted between 48-72h after the deadline, a 15 percentage point late penalty will be applied to the grade. For homeworks submitted 72h or more after the deadline, a 100 percentage point late penalty will be applied to the grade.

For example, if a student submits their homework 25h after the deadline and receive a pre-penalty grade of 87%. Their grade with the late penalty incorporated would be $87\% - 5\% = 82\%$.

Each student starts the course with two "72h of no late penalties" coupons, each of which allows them to choose one homework from the course and remove the 5, 10, or 15 percentage point late penalty (whichever one is applicable). Importantly, if you hand in the assignment 72h or more passed the deadline, you will still receive a zero on the assignment. In order to use your

free pass, simply email me and the GSI (in one email) and say that you would like to use your free pass for a homework.

Collaboration and Plagiarism

You must write your homework submission on your own and in your own words. If you worked with other students on the homework, you must list all of them. Excessive reliance on other students or large language models won't help you in the long run. Imagine you used it throughout elementary school, where would you be now?

See the MSI Student Handbook for the definition of Plagiarism to help avoid it.

Accommodations for Students with Disabilities

The University of Michigan recognizes disability as an integral part of diversity and is committed to creating an inclusive and equitable educational environment for students with disabilities. Students who are experiencing a disability-related barrier should contact [Services for Students with Disabilities \(Links to an external site.\)](https://ssd.umich.edu/) <https://ssd.umich.edu/>; 734-763-3000 or ssdoffice@umich.edu). For students who are connected with SSD, accommodation requests can be made in Accommodate. If you have any questions or concerns please contact your SSD Coordinator or visit SSD's Current Student webpage. SSD considers aspects of the course design, course learning objects and the individual academic and course barriers experienced by the student. Further conversation with SSD, instructors, and the student may be warranted to ensure an accessible course experience. The instructional team will treat any information that you provide in as confidential a manner as possible.

More broadly, please come to the teaching team with any issues you have that are hindering your success in the classroom. Learners face a variety of physical, cognitive, and emotional challenges, and we often have experience in how to accommodate those challenges for different types of assignments. We prefer to work with you to come up with a mutually agreeable plan for accommodation, and are happy to consider any accommodation that increases your ability to learn.

Student Mental Health and Wellbeing:

University Students may experience stressors that can impact both their academic experience and their personal well-being. These may include academic pressures and challenges associated with relationships, mental health, alcohol or other drugs, identities, finances, etc. If you are experiencing concerns, seeking help is a courageous thing to do for yourself and those who care about you. If the source of your stressors is academic, please contact me so that we can find solutions together. Ashley Ewearitt, a Counseling and Psychological Services (CAPS) counselor, is embedded in UMSI, schedule an appointment with her by copying and pasting the following url into your browser: <https://bit.ly/Ewearitt-CAPS-UMSI> . For personal concerns, U-M offers a variety of resources, many which are listed on the [Health and Wellness](#) webpage.